BCS358C – Git Lab
Program 5: Collaboration and Remote Repositories

Aim: To fetch the latest changes from a remote repository and rebase the local branch onto the updated remote branch.

Main commands in the syllabus: git clone, git pull, git push, and git rebase.

Commands and Steps

1. Clone the remote repository (run once)
git clone <repository-URL>

Downloads a copy of the remote repository to your computer. Replace <repository-URL> with the actual repository URL.

2. Move into the project folder
cd <repository-folder>

Changes the current directory to the downloaded project folder.

3. Check the current repository status (basic supporting command)
git status

Shows the current branch and whether any files have been changed.

4. Fetch the latest changes and rebase your branch
git pull --rebase origin main

Fetches the latest changes from the remote main branch and replays your local commits on top of the updated branch. This is the key command for the stated objective.

5. Stage your changes (only if you edited or created a file)
git add <filename>

Stages the file so it can be included in a commit. Use the actual file name, for example: git add file.txt.

6. Commit your changes (only if you staged changes)
git commit -m "Updated file"

Saves your staged changes as a local commit.

7. Push local commits to the remote repository (if you have commits to share)
git push origin main

Uploads your local commits to the remote main branch.

NOTE : If a Rebase Conflict Occurs

If Git reports a conflict, open the conflicted file and resolve the marked sections. Then run:

git add <filename>
git rebase --continue

To cancel the rebase and return to the state before it started, run:

git rebase --abort

Important Notes

- Use the correct remote branch name. The examples use main; some repositories use master or another branch name.
- Run git clone only when creating your local copy for the first time.
- For this program, use git pull --rebase rather than running git pull origin main and then git pull --rebase origin main. The two pulls in sequence are unnecessary.
- The add, commit, and push commands are needed only when you have local changes to save and share. If the task is only to update your local branch, the key command is git pull --rebase origin main.

Short Command Summary

- git clone <repository-URL> — copy the remote repository locally
- cd <repository-folder> — enter the project folder
- git status — check the working tree and branch
- git pull --rebase origin main — fetch remote updates and rebase local commits
- git add <filename> — stage a changed file
- git commit -m "Updated file" — save staged changes in a commit
- git push origin main — upload local commits
- git rebase --continue — continue after resolving a conflict
- git rebase --abort — cancel the rebase